

Yuhang Qiu

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CONTACT INFORMATION

Email: qiuyuhang@sjtu.edu.cn

EDUCATION

Shanghai Jiao Tong University

B.S. in Computer Science

Undergraduate Researcher, X-LANCE Lab, School of Computer Science

Advisor: Prof. Kai Yu

Member of ACM Honors Class, an elite CS program for top 5% students

Sep. 2023 -- Present

Expected Graduation: Jun. 2027

GPA: **3.91**/4.3

Selected Courses:

- Computer Vision: **A+**, **100/100**
- Principle and Practice of Computer Algorithms: **A+**, **99/100**
- Optimization Methods: **A+**, **98/100**
- Linear Algebra (Honor): **A+**, **97/100**
- Compiler Design and Implementation: **A+**, **95/100**
- Large Language Models: **A+**, **95/100**
- Computer Networks: **A+**, **95/100**
- Introduction to Physics (H) II: **A+**, **95/100**

RESEARCH INTERESTS

Speech AI, with a focus on speech understanding and generation. I am particularly interested in the reliability and robustness of Automatic Speech Recognition (ASR), Large Audio Language Models (LALMs), and ASR--TTS co-evolutionary systems.

RESEARCH EXPERIENCE

Undergraduate Researcher, X-LANCE Lab, Shanghai Jiao Tong University

Advisor: Prof. Kai Yu

Jun. 2025 -- Present

• An ASR-TTS Collaborative Training Framework

Mar. 2026 -- Present

- Developing an iterative ASR-TTS collaborative training framework to couple speech recognition and speech generation through mutual data selection.
- Designing a training loop in which each model serves as a reward model for the other to guide data filtering, with the selection strategy updated throughout training.

• Reliable Automatic Speech Recognition

Oct. 2025 -- Mar. 2026

- Conducting research on reliable ASR, aiming to enable models to detect potential recognition errors and explicitly abstain at the word level.
- Developing evaluation metrics that jointly consider accuracy and precision, and designing methods to improve precision while preserving overall recognition performance.

• ISA-Bench: Instruction Sensitivity Benchmark

Jun. 2025 -- Sep. 2025

- Contributed to ISA-Bench, a dynamic benchmark evaluating instruction sensitivity of Large Audio Language Models along instruction description, output format, and task composition.
- Conducted empirical analysis showing that state-of-the-art LALMs exhibit significant instruction sensitivity, motivating improvements in model robustness and reliability.

PUBLICATIONS

Bohan Li*, Wenbin Huang*, **Yuhang Qiu***, *et al.* *ISA-Bench: Benchmarking Instruction Sensitivity for Large Audio Language Models* [arXiv preprint arXiv:2510.23558](#), 2025. [Accepted to *ICASSP 2026*] (* equal contribution)

Wenbin Huang*, **Yuhang Qiu***, *et al.* *RAS: a Reliability Oriented Metric for Automatic Speech Recognition* [arXiv preprint arXiv:2604.24278](#), 2026. [Under review at *Interspeech 2026*] (* equal contribution)

PROJECTS

MxCompiler

[Link to Repo](#)

A course project of Compiler Design and Implementation

It's a compiler of Mx*(a C-like language for education), written in Java.

VerilogCPU

[Link to Repo](#)

A course project of Computer Architecture.

It's a toy CPU written in Verilog.

- **Tomasulo algorithm** is implemented.
- Basic instructions in RISC-V are supported.

AWARDS AND HONORS

2025 Zhiyuan Honors Scholarship (2 % in SJTU)

2025 Academic Excellence Scholarship

2025 Excellent Teaching Assistant (10 % TAs)

2024 Academic Excellence Scholarship

2024 Zhiyuan Honors Scholarship (2 % in SJTU)

2023 Zhiyuan Honors Scholarship (2 % in SJTU)

TEACHING EXPERIENCE

Programming, Teaching Assistant 2024, Fall

Data Structure, Teaching Assistant 2025, Spring

SKILLS

Programming C++, Python, Go, Java, Verilog

Tools PyTorch, Git, Markdown, LaTeX

Languages Chinese(Native), English(Fluent)